Why the SK order parameter has no gaps

A variational argument from contact to strict curvature

Expository draft, 26 October 2026

The Sherrington–Kirkpatrick (SK) model asks us to choose signs that maximise a random sum of pair interactions. The Parisi formula expresses the limiting maximum through a nondecreasing function, the **order parameter**. We explain why a minimising order parameter cannot be constant on any interval below one.

The proof compares two consequences of a hypothetical gap. Minimality fixes the values and bounds the slopes of an auxiliary curve at its endpoints. An analytic estimate makes the derivative strictly convex between them, leaving too little area to connect the required values. Initial and terminal gaps need separate arguments. We then return to the spins: a theorem about their energies gives nearly optimal sign choices differing on any prescribed positive fraction of positions up to one half, with arbitrarily small error in that fraction.

We follow the quantitative proof in OpenAI (2026a); Hong-Bin Chen (2026), Theorem 1.3 had already proved full support by a different method. The estimate used here makes the final contradiction a direct comparison of areas. We assume algebra and elementary calculus, and introduce probability and cumulative mass through calculations. The deeper analytic facts are stated where the proof needs them. By the end of Sections 1–6, the reader should be able to reconstruct how minimality and those inputs exclude each kind of gap, then translate the result into a statement about spin configurations. Section 7 gives optional calculations and answers to the short questions in the text; the full analytic proofs remain in the cited sources.

1. The finite optimisation problem

At each of n positions, or **sites**, we choose a sign $\sigma_i \in \{-1, 1\}$, called a **spin**. A configuration is the vector $\sigma = (\sigma_1, \dots, \sigma_n)$. For each unordered pair choose an independent standard Gaussian J_{ij}, a normal random variable with mean zero and variance one. Independence means that learning the other couplings does not change the probability distribution of this one. These coefficients, or **couplings**, are drawn once and then held fixed while we maximise the energy, or **Hamiltonian**,

$$H_n(\sigma) = \frac{1}{\sqrt{n}} \sum_{1 \le i < j \le n} J_{ij}\sigma_i\sigma_j.$$

A positive coupling rewards equal signs; a negative one rewards opposite signs. For example, positive couplings on pairs 12 and 23 favour $\sigma_1 = \sigma_2 = \sigma_3$, which conflicts with a negative coupling on pair 13. This inability to maximise every pair's contribution simultaneously is called **frustration**. The factor $n^{-1/2}$ fixes the model's scale.

For two spins, $H_2(\sigma) = J_{12}\sigma_1\sigma_2/\sqrt{2}$. Choosing equal or opposite signs according to the sign of J_{12} gives $\max_\sigma H_2(\sigma) = |J_{12}|/\sqrt{2}$.

To average over the couplings, we first need their probabilities. A **probability density** assigns probability by area. For a standard Gaussian random variable Z, the density is $\varphi(z) = (2\pi)^{-1/2}e^{-z^2/2}$; it is nonnegative, has total integral one, and for $a \le b$ gives

$$\Pr(a \le Z \le b) = \int_a^b \varphi(z)\,dz.$$

In particular, a single point has probability zero. The density's height is not the probability of that point. To average a score $f(Z)$, weight it by this same density. Its **expectation** is

$$\mathbb{E}f(Z) = \int_{\mathbb{R}} f(z)\varphi(z)\,dz.$$

For the standard Gaussian, $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = 1$. For a mean-zero random variable, its **variance** is this mean square; thus $\sqrt{h}Z$ has variance h for $h > 0$. Symmetry gives the zero mean, whereas $\mathbb{E}|Z| = 2\int_0^\infty z\varphi(z)\,dz = \sqrt{2/\pi}$. Thus the expected two-spin maximum is $1/\sqrt{\pi}$. For any signs chosen before seeing the coupling, the expected energy is instead zero. The order matters: we choose signs for each realised coupling, then average the achieved maximum. For many spins, we seek the limiting expected maximum **per spin**,

$$P_* := \lim_{n\to\infty} \frac{1}{n}\mathbb{E}\max_{\sigma} H_n(\sigma),$$

where expectation averages over the couplings after taking the maximum. **Zero temperature**, or **ground-state value**, refers here to this maximisation problem. The model is **pure** because every term contains exactly two spins, and has **zero external field** because there is no additional term $h\sum_i \sigma_i$.

The maximum is one question; the arrangement of nearly optimal choices is another. To compare two configurations for the *same couplings*, define their **overlap** $R(\sigma,\tau) = n^{-1}\sum_i \sigma_i\tau_i$, their average agreement with a disagreement counted as -1. If they disagree at k positions, then $R = 1 - 2k/n$. For example, $(+,+,+,+)$ and $(+,+,+,-)$ have overlap $1/2$; two disagreements would give zero. We will ask whether two nearly optimal configurations can differ on roughly one quarter of their sites, and more generally at any prescribed fraction up to one half.

Our normalisation gives the covariance function $\xi(t) = t^2/2$ in the variational problem below. **Optional normalisation calculation.** For fixed σ, τ, put $v_i = \sigma_i\tau_i$. When we expand the product of energies, different couplings have mean product zero by independence and their zero means; a coupling multiplied by itself has mean one. Together with $v_i^2 = 1$, this gives

$$\mathbb{E}[H_n(\sigma)H_n(\tau)] = \frac{1}{n}\sum_{i<j} v_i v_j = \frac{(\sum_i v_i)^2 - n}{2n} = \frac{n}{2}R(\sigma,\tau)^2 - \frac{1}{2}.$$

Dividing by n gives the limiting covariance $\xi(t) = t^2/2$. For exact covariance $n\xi(R)$, add the same independent centred Gaussian of variance $1/2$ to every energy; this leaves the expected maximum unchanged.

Back to the spin question. Reversing one configuration's signs preserves its energy and reverses its overlap with another. We next define deterministic candidate values indexed by nonnegative overlap scales t. The Parisi theorem in Section 3 identifies their minimum with the limiting spin optimum; that connection is an external input, not a change of variables in the finite maximisation.

2. What a gap means

A candidate order parameter records accumulated weight. Begin with a finite example: place weight two at scale $1/4$ and weight one at scale $3/4$. The cumulative function is zero, then two, then three:

$$\gamma(t) = \begin{cases} 0, & 0 \le t < 1/4, \\ 2, & 1/4 \le t < 3/4, \\ 3, & 3/4 \le t < 1. \end{cases}$$

Its area is $2(1/2) + 3(1/4) = 7/4$. We can count the same area by following each weight: the first persists for three quarters of the interval and the second for one quarter, giving $2(3/4)+1(1/4) = 7/4$ again. Write δ_s for one unit of mass at s and $\mu = 2\delta_{1/4} + \delta_{3/4}$ for these weights. Integration against this measure is a weighted sum:

$$\int F(s)\,\mu(ds) = 2F(1/4) + F(3/4).$$

The weights total three, so this integral is not a probability average. We will reuse this accounting when each weight has a cost.

In general, an admissible **order parameter** is a function $\gamma : [0,1) \to [0,\infty)$ that is nondecreasing, right-continuous and has finite integral. Right-continuity includes a jump's new value at the scale where the jump occurs, as in the example. The associated **Stieltjes measure** records the increments:

$$\mu_\gamma([0,t]) = \gamma(t), \qquad \mu_\gamma((a,b]) = \gamma(b) - \gamma(a).$$

The first equation permits mass $\gamma(0)$ at zero. Mass concentrated at one point is an **atom**. The weights need not be probabilities of finite-system overlaps, and their total need not be one or even finite. They are finite on each $[0,T]$ with $T < 1$. Admissibility requires instead the weighted finiteness

$$\int_0^1 \gamma(t)\,dt = \int_{[0,1)} (1-s)\,\mu_\gamma(ds) < \infty.$$

This is the same two ways of counting area as above: a unit at s contributes at every later time, for a total length $1-s$. The general identity uses the integration theorem that nonnegative contributions may be summed in either order. A mass at one would change none of the cumulative values below one, so one is excluded from the domain.

The **support** consists of points whose every neighbourhood carries positive mass. A neighbourhood here is an interval around the point, clipped to $[0,1)$. Full support means that every nonempty relative open interval carries mass. Equivalently, γ is strictly increasing: full support gives a positive increment on each $(a,b]$ with $a < b$, and a positive increment on a smaller interval inside any neighbourhood gives positive mass there.

For example, the admissible candidate

$$\gamma(t) = \begin{cases} t, & 0 \le t \le 1/3, \\ 1/3, & 1/3 \le t \le 2/3, \\ t - 1/3, & 2/3 \le t < 1 \end{cases}$$

has zero increment from 2/5 to 3/5, but increment 1/8 from 1/8 to 1/4. Its measure has density one outside $(1/3, 2/3)$ and zero inside: the mass of a set is the integral of this density there. Both endpoints of the gap remain in the support because every neighbourhood of either reaches a region carrying positive mass. The upper curve in Figure 1 stays positive across the gap; it is its *increments* that vanish. This candidate has infinitely many support points, including points arbitrarily close to one, yet still has a middle gap. Those two weaker properties do not give full support.

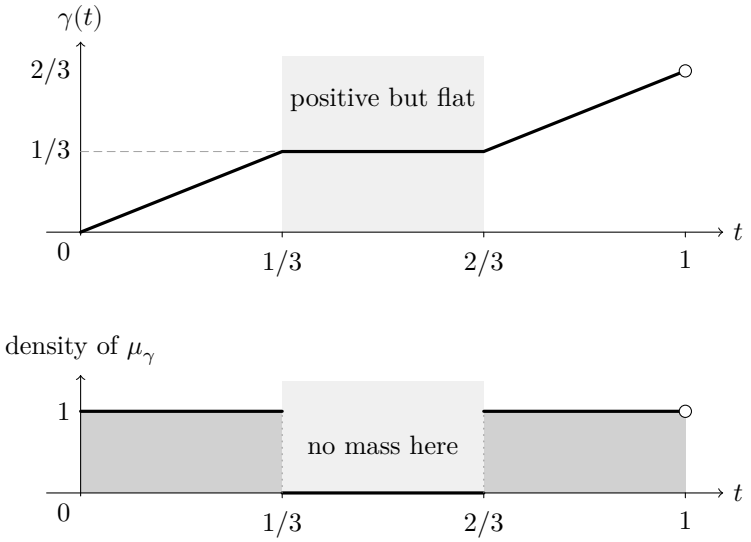


Figure 1: The same candidate in two representations. The upper curve is the piecewise-linear γ from Section 2; the lower plot shows the density of its Stieltjes measure. On $(1/3, 2/3)$ the cumulative value stays positive while its increments vanish. The endpoints 1/3 and 2/3 still belong to the support: every neighbourhood of either contains positive mass. The lower density belongs to this example; a general Stieltjes measure need not have one. Open circles mark the excluded endpoint 1. The candidate is not a computed minimiser. Axes and guides adapt OpenAI (2026a), Figure 1; the paired example is constructed here.

A constant candidate $\gamma(t) = c > 0$ makes the distinction sharper: its measure is $c\delta_0$, supported only at zero. Positivity of the function is not the conclusion we seek. Nor does full support alone assert a density: a measure with summable positive masses at a dense countable set has full support. The further-reading note returns to the stronger density conclusion.

Question 1: allowed changes. Return to the two weights $2\delta_{1/4} + \delta_{3/4}$. For $0 < \varepsilon < 1$, which changes leave an admissible cumulative function: adding ε mass at 1/2, removing it at 1/4, removing it at 1/2, or multiplying both weights by $1 - \varepsilon$? Check monotonicity as well as nonnegativity. Answers are in Section 7.4.

3. How a candidate receives its value

To rule out a candidate, we will look for an allowed change that lowers its value. Adding mass at one scale changes the cumulative coefficient at every later scale. We therefore need both a value for each candidate and a way to compute the cost of that change. The auxiliary optimisation below supplies the value; its response to the starting state will enter the first-variation formula in Section 5.1.

3.1 First evaluate the zero coefficient

The variational problem assigns each γ a value by a one-dimensional auxiliary optimisation. Its final reward at state x is $|x|$: this is the best one-spin score $\max_{\sigma\in\{-1,1\}} x\sigma$. First take $\gamma = 0$, when only noise changes the state. The auxiliary clock is chosen so that noise accumulated over an interval of length h has variance h, the variance of $\sqrt{h}Z$. Starting at time t in state x, the remaining noise is therefore distributed as $\sqrt{1-t}Z$. The expected final reward is

$$\Phi_0(t,x) = \mathbb{E}|x + \sqrt{1-t}Z| = \int_{\mathbb{R}} |x + \sqrt{1-t}z|\varphi(z)\,dz.$$

At $(t,x) = (0,0)$ this is the calculation already performed: $\Phi_0(0,0) = \sqrt{2/\pi}$. At $t = 1$ no noise remains and the value is $|x|$. The function gives an expected payoff at a specified starting state; it is not a probability density.

3.2 Allowing a response to the noise

For a general γ, a response can bias the future state, with a quadratic cost. Here the overlap scale t plays the role of time; the path is auxiliary, not a motion of the spins. The noise is **Brownian motion** B: a continuous random path starting at zero. Its increments over disjoint time intervals are independent; over an interval of length h, an increment has the same distribution as the centred Gaussian $\sqrt{h}Z$.

A **control** $\alpha_s \in [-1,1]$ chooses a response at each time using only noise already observed since the starting time. It can react to the past but cannot use future increments. To see why this matters, let a sign $W = \pm 1$, each with probability $1/2$, be revealed after a single choice $v \in [-1,1]$. The expected score $|W+v| - v^2/2$ is $1 - v^2/2$, maximised at $v = 0$. A choice $v = W$ would score $3/2$, but uses information unavailable at the time of choice. The continuous control problem repeats decisions with this same restriction; the finite example only illustrates that restriction.

For a starting position x, the value is

$$\Phi_\gamma(t,x) = \sup_\alpha \mathbb{E}\left[\left|x + B_1 - B_t + \int_t^1 \gamma(s)\alpha_s\,ds\right| - \frac{1}{2}\int_t^1 \gamma(s)\alpha_s^2\,ds\right].$$

Inside the absolute value, noise and the accumulated bias change the state before its final reward is paid. The second integral is the cost. The supremum chooses the best permitted rule; its formal information condition is progressive measurability. Expectation now averages over Brownian paths for a fixed γ, rather than over spin couplings. At $\gamma = 0$ both control terms vanish, recovering Section 3.1.

The **Parisi functional** subtracts an explicit correction from this optimised value:

$$\mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \frac{1}{2}\int_0^1 t\gamma(t)\,dt.$$

The inner supremum chooses a response for fixed γ; the outer problem minimises $\mathcal{P}$ over γ. The ground-state Parisi formula and attainment give

$$P_* = \min_{\gamma \text{ admissible}} \mathcal{P}(\gamma).$$

Each candidate supplies an upper bound, and minimisation finds the sharpest one. This equality and attainment are external inputs from Auffinger–Chen (2017), Theorem 1 and the proof of Lemma 3; Gaussian concentration gives the expected-value form used here. The no-gap theorem assumes a minimiser and proves a property of it. The value $\mathcal{P}(0) = \sqrt{2/\pi}$ alone does not establish that the zero candidate minimises.

3.3 The response, the state and the averaged square

First consider the value as a function of the starting state, holding t and γ fixed; we also write $\Phi = \Phi_\gamma$. For any permitted control, use the same responses to each Brownian history at two starting states x and y. The costs are unchanged, and the triangle inequality bounds the change in the terminal reward by $|x - y|$. Taking the supremum over controls gives

$$|\Phi_\gamma(t,x) - \Phi_\gamma(t,y)| \le |x - y|.$$

The value is also **convex**: between two states its graph lies at or below the chord joining their values. Indeed, each fixed-control payoff is an average of translated absolute-value functions, hence convex. Its chord lies at or below the chord of the best endpoint payoffs; taking the supremum preserves that upper bound.

OpenAI (2026a), Lemmas 2.1–2.2 supply spatial differentiability below time one. Let $u(t,x) = \partial_x \Phi_\gamma(t,x)$ be the slope as the starting state changes. A small shift δ changes the best expected payoff by approximately $u(t,x)\delta$. Write $a(t,x) = \partial_{xx}\Phi_\gamma(t,x)$ for the rate at which that slope changes. The displayed bound gives $|u| \le 1$, and convexity gives $a \ge 0$. These are spatial derivatives: time is held fixed.

For the zero coefficient, the slope can be computed directly. Put $F_{\mathrm{G}}(z) = \Pr(Z \le z) = \int_{-\infty}^{z} \varphi(y)\,dy$. Moving x slightly raises $|x + \sqrt{1-t}Z|$ when the final state is positive and lowers it when the final state is negative. Consequently

$$u_0(t,x) = 2F_{\mathrm{G}}\left(\frac{x}{\sqrt{1-t}}\right) - 1.$$

Differentiating the integral is legitimate: the difference quotients of the absolute value are bounded by one, and the Gaussian has zero probability of hitting its corner. Thus the slope is the probability of a positive final state minus the probability of a negative one.

For general γ, the control theorem says that the slope response attains the value. Its local optimisation has a simple algebraic reason: at slope u, the gain from a response v minus its quadratic cost is

$$\gamma vu - \tfrac{1}{2}\gamma v^2 = \tfrac{1}{2}\gamma u^2 - \tfrac{1}{2}\gamma(v-u)^2.$$

Since $\gamma \geq 0$ and $|u| \leq 1$, $v = u$ attains its maximum. The optimised contribution $\gamma u^2/2$ also explains why a squared slope will enter the sensitivity to γ. The control theorem and the first-variation formula in Section 5.1 make these connections rigorous; the square completion alone does not justify differentiating an optimised value. The zero-temperature control representation is in Auffinger–Chen, Corollary 2, equation (12), and the extension after (13).

Following the slope response generates the random state

$$X_t = B_t + \int_0^t \gamma(s)u(s, X_s)\,ds, \qquad X_0 = 0.$$

This is the **associated diffusion**. The rule is called **feedback** because it uses the slope at the currently reached state. With γ and u supplied, a short forward step of length h from state x at time t, with $t + h < 1$, is approximated by

$$x + h\gamma(t)u(t, x) + \sqrt{h}Z.$$

The response is evaluated before the fresh independent noise Z is observed. This discrete approximation explains the order of operations; the integral equation defines the continuous process. Its drift accumulates through time, so even a jump of γ does not make X jump.

Fix γ, follow a path to X_t, evaluate the slope there, square it, and average over paths. This defines a deterministic curve

$$q(t) := \mathbb{E}[u(t, X_t)^2].$$

Here Φ_γ and u are deterministic functions; X_t and the response evaluated at it are random. Squaring before averaging retains the size of both positive and negative responses.

A calculation at half time. For $\gamma = 0$, $X_t = B_t$ and $B_{1/2}$ has the same distribution as $Z/\sqrt{2}$. We can average over paths by grouping them according to the state reached. At half time the reached state has density $e^{-x^2}/\sqrt{\pi}$. This density describes its **law**, assigning probability to positions on the real line. The law has total mass one even though the order-parameter measure μ_0 is zero. The two measures describe different objects.

Average $u_0(1/2, x)^2$ with this density. Substituting $x = z/\sqrt{2}$ gives

$$\begin{aligned} q_0(1/2) &= \int_{\mathbb{R}} (2F_{\mathrm{G}}(z) - 1)^2 \varphi(z)\,dz \\ &= \int_0^1 (2v-1)^2\,dv = \frac{1}{3}, \qquad v = F_{\mathrm{G}}(z), \quad dv = \varphi(z)\,dz. \end{aligned}$$

The response at zero is zero, but its mean square at the random state is $1/3$. Thus $q(t) = t$ is not automatic. Although every control is optimal at $\gamma = 0$, q_0 uses the value slope u_0. Section 7.1 uses the whole curve to find a feasible decrease of $\mathcal{P}(0)$; a discrepancy at one time does not suffice.

Question 2: change the time. Still take $\gamma = 0$ everywhere, but now set $t = 1/4$. What variance has the state accumulated, and what variance remains in its future reward? Write the Gaussian integral for $q_0(1/4)$ without evaluating it. Section 7.4 gives the answer.

The analytic facts needed in the proof are

$$q \in C^1([0,1)), \qquad q(0) = 0, \qquad 0 \le q(t) \le 1, \qquad q'(t) = \mathbb{E}[a(t, X_t)^2].$$

Here C^1 means continuously differentiable, with derivatives at zero from the right. In particular, q' stays continuous across jumps of γ. Spatial curvature controls the growth of the squared response. These facts hold for every admissible candidate and follow from OpenAI (2026a), Lemmas 2.1–2.2. They suffice for the proof below together with the further inputs stated at their uses. Section 7.2 explains the PDE and stochastic calculation behind them. The next argument uses these stated properties of the deterministic curve q to test changes of the cumulative weight γ.

4. The full-support theorem

Source theorem (OpenAI (2026a), Theorem 1.1). In the pure, zero-field model with $\xi(t) = t^2/2$, let γ be an admissible integrable minimiser of $\mathcal{P}$. Its Stieltjes measure has full relative support on $[0, 1)$, and γ is strictly increasing. For every associated diffusion and every $0 \le t < 1$,

$$\mathbb{E}[u(t, X_t)^2] = t, \qquad \mathbb{E}[a(t, X_t)^2] = 1.$$

The cumulative function determines its measure uniquely; for a fixed Brownian motion the associated diffusion is unique by the cited regularity results. The theorem requires a minimiser, not merely an admissible candidate. Zero field supplies $q(0) = 0$; $\xi'' = 1$ enters the curvature estimate. The statement does not cover nonzero field or general mixed-spin covariance functions.

5. Why a flat interval cannot minimise

5.1 Allowed changes and their first-order cost

The **first variation** in a direction η is the derivative at zero of $\varepsilon \mapsto \mathcal{P}(\gamma + \varepsilon\eta)$. At a minimum its right derivative is nonnegative whenever small $\varepsilon \ge 0$ leave the candidate admissible. It need not vanish: the opposite change may violate nonnegativity or monotonicity. Precisely, let $\eta \in L^1(0,1)$, meaning $\int_0^1 |\eta(t)|\, dt < \infty$, and suppose $\gamma + \varepsilon\eta$ stays admissible for every $0 \le \varepsilon \le \varepsilon_0$, for some $\varepsilon_0 > 0$. The formula is

$$D\mathcal{P}(\gamma)[\eta] = \frac{1}{2} \int_0^1 \eta(t)(q(t) - t)\, dt.$$

Here q belongs to the unperturbed candidate γ. The term $-t/2$ comes from the explicit penalty in $\mathcal{P}$; $q(t)/2$ is the sensitivity of the reoptimised control value. The latter is the analytic content of the formula, not differentiation with a fixed control. Chen–Handschy–Lerman (2018), Proposition 4 gives the derivative along the segment joining any two admissible coefficients. Taking the second endpoint to be $\gamma + \varepsilon_0\eta$ and rescaling the segment parameter gives the formula above. The detailed proof in OpenAI (2026a), Lemma 6.1 compares value functions and bounds the terminal remainder by the L^1 tail of η.

Adding a small atom at scale s raises the cumulative function at *every* later scale. It is always allowed: the total mass is not fixed, so no compensating removal is required. Its direction is $\eta(t) = \mathbf{1}_{[s,1)}(t)$, equal to zero before s and one from s onward. Thus

$$D\mathcal{P}(\gamma)[\mathbf{1}_{[s,1)}] = \frac{1}{2}\int_s^1 (q(t) - t)\,dt =: \frac{1}{2}G(s).$$

For the candidate in Figure 1, adding ε mass at $s = 1/2$ creates a jump of height ε there and raises the entire remaining curve by ε. The coefficient changes only on that tail, but its optimised value and feedback must be recomputed: earlier starting states also face a changed future control problem. The first-variation theorem accounts for this dependence. Its formula has a tail integral because that is where the coefficient changes: $G(s)/2$ is the marginal cost of placing mass at s. At a minimiser, $G(s) \geq 0$ for every $s < 1$.

For the two weights in Section 2, the same accounting gives

$$\int_0^1 \gamma(t)(q(t) - t)\,dt = 2G(1/4) + G(3/4).$$

Each weight is charged for its entire remaining tail. The q and G in this identity belong to that candidate; no optimality has been assumed. Reducing both weights by a fraction ε multiplies the cumulative function by $1 - \varepsilon$.

Suppose temporarily that this two-weight candidate minimised $\mathcal{P}$. Addition would give two non-negative costs, $G(1/4) \geq 0$ and $G(3/4) \geq 0$, while the allowed reduction would give

$$0 \leq -\frac{1}{2}(2G(1/4) + G(3/4)).$$

Both costs must be zero: their positive-weighted sum is nonnegative, and the reduction forces it to be nonpositive. This is a necessary condition under the temporary minimiser assumption, not a claim that the two-weight candidate attains the minimum.

For a general minimiser, the same argument replaces the finite sum by an integral. The direction $\eta = -\gamma$ is integrable, and $(1 - \varepsilon)\gamma$ remains admissible for $0 \leq \varepsilon \leq 1$. It scales all existing mass by the same factor. For the continuous candidate in Figure 1, it multiplies the two nonzero slopes and the plateau height by $1 - \varepsilon$; it works for diffuse mass as well as atoms. The weighted-sum identity becomes

$$0 \leq D\mathcal{P}(\gamma)[-\gamma] = -\frac{1}{2}\int_0^1 \gamma(t)(q(t) - t)\,dt = -\frac{1}{2}\int_{[0,1)} G(s)\,\mu_\gamma(ds).$$

The exchange of integration order is valid even if the total mass is infinite. Its absolute integral is bounded by the area from Section 2:

$$\int_{[0,1)}\int_s^1 |q(t) - t|\,dt\,\mu_\gamma(ds) = \int_0^1 \gamma(t)|q(t) - t|\,dt \leq \int_0^1 \gamma(t)\,dt < \infty.$$

Since $G \geq 0$, its integral must be zero. If G were positive at a support point, continuity would bound it below by a positive number on a neighbourhood carrying positive mass, contradicting that zero integral. Hence $G = 0$ on the support. This is **contact**: the nonnegative function touches zero there. The scaling argument is the variational argument of Chen, Proposition 3.2, equations (3.11)–(3.12), applied with the integrable-direction formula above.

Scaling removes only mass that exists. In Figure 1, even the supported point $1/4$ has no atom: intervals around it carry mass, but the point itself does not. Subtracting $\varepsilon\delta_{1/4}$ would create a downward jump. For $0 < \varepsilon < 1/4$ the cumulative values would remain nonnegative, but the function would no longer be nondecreasing. Support alone does not permit atom removal. We have proved contact on the support, not that every zero of G must carry mass.

5.2 Contact gives endpoint conditions

We need information about q at the ends of a possible gap. An interior support point s is a minimum of G. Since $G'(s) = s - q(s)$ and $G''(s) = 1 - q'(s)$, ordinary calculus gives

$$q(s) = s, \qquad q'(s) \leq 1.$$

At zero, if it belongs to the support, contact and symmetry give $G(0) = q(0) = 0$. If $q'(0) > 1$, continuity would give $q'(t) > 1$ for small positive t. Integration would then give $q(t) > t$, so $G'(t) = t - q(t) < 0$. Starting from $G(0) = 0$, this would force $G(t) < 0$, a contradiction. Hence $q'(0) \leq 1$. This one-sided argument includes an atom at zero.

5.3 Strict curvature creates an impossible integral

The support is relatively closed in $[0, 1)$. Starting at a missing point, enlarge its mass-free interval until it reaches support or a boundary of the domain. Such a maximal interval has two support endpoints, starts at zero outside the support, or reaches one. We treat these three cases in order; the last includes empty support.

First suppose (r, s) has both endpoints in the support, with $0 \leq r < s < 1$. Its constant height c is necessarily positive: for t inside the gap, a small relative neighbourhood of r lies in $[0, t]$ and has positive mass, so $c = \mu_\gamma([0, t]) > 0$. If $r = 0$, this mass is an atom at zero. Contact imposes

$$q(r) = r, \quad q(s) = s, \quad q'(r) \leq 1, \quad q'(s) \leq 1.$$

The first two conditions require area $s - r$ under q' across the gap. A calculus model shows how strict convexity can prevent this. On the earlier interval $[1/3, 2/3]$, put $x = 3t - 1$ and consider the slope $1 - x(1 - x)$. Its endpoints equal one, but

$$\int_{1/3}^{2/3} [1 - (3t - 1)(2 - 3t)]\, dt = \frac{1}{3} \int_0^1 [1 - x(1 - x)]\, dx = \frac{5}{18}.$$

The missing area is $1/3 - 5/18 = 1/18$. This is a calculus model, not a computed SK moment curve. The next input forces the actual q' below its endpoint chord in the same way.

Strict-curvature input (OpenAI (2026a), Proposition 5.1). For any admissible coefficient equal to $c > 0$ on an open interval, its moment curve satisfies, strictly inside that interval,

$$q'''(t) \geq \frac{c^2}{1000}\mathbb{E}[a(t, X_t)^4] > 0 \qquad (r < t < s).$$

Section 7.3 explains how this sign is obtained after expectation. The gap argument uses its strict sign; it does not use the particular number $1/1000$. The final inequality uses $a(t, x) > 0$, proved by Gaussian smoothing on the constant interval (OpenAI (2026a), Lemma 2.3); convexity of Φ alone would only give $a \geq 0$. Because $q''' = (q')''$, the function q' is strictly convex: between two points its graph lies strictly below their joining chord. The moment identity makes q' continuous at the endpoints, even if γ jumps there; higher derivatives are needed only inside the gap. Hence

$$q'(t) < \frac{s-t}{s-r}q'(r) + \frac{t-r}{s-r}q'(s) \leq 1 \qquad (r < t < s).$$

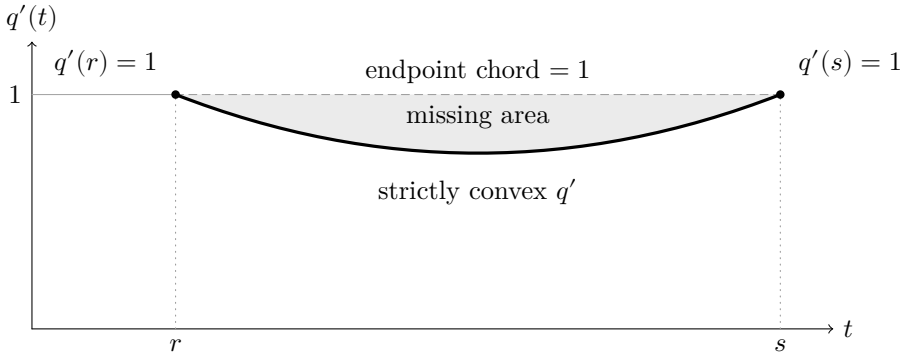


Figure 2: The case where strictness is indispensable: both endpoint slopes equal one, so the chord alone loses no area. A strictly convex slope lies below it. Contact would require zero shaded area, but $\int_r^s (1 - q'(t))\,dt > 0$. The drawn calculus model is $1 - x(1-x)$ with $x = (t-r)/(s-r)$; its mean is $5/6$, while the constant slope one has mean one. Neither is asserted to be an SK moment curve. The proof also allows unequal endpoint slopes, each at most one. Adapted from Figure 1 of OpenAI (2026a), OpenAI/math (Apache 2.0).

Integration gives $q(s) - q(r) < s - r$, contradicting contact. For the candidate in Section 2, a minimising plateau on $(1/3, 2/3)$ would require the same increment to be both equal to and strictly less than $1/3$.

5.4 The remaining initial and terminal gaps

Initial gap with zero outside the support. Suppose $[0, s)$ is missing and $s < 1$ is the first support point. Then $\gamma = 0$ there, so the positive-coefficient bound supplies no strictness. Instead the zero-coefficient heat identity gives

$$q''(t) = \mathbb{E}[\Phi_{xxx}(t, X_t)^2] > 0 \qquad (0 < t < s).$$

Here is why the inequality is strict. On this interval $X_t = B_t$, whose density is positive everywhere at positive times. If the expected square were zero, continuity would force $\Phi_{xxx}(t, x) = 0$ for every x: a nonzero value would give a small interval on which the square is positive, and that

interval has positive probability. Then $\Phi(t,\cdot)$ would be a quadratic polynomial, and its bounded slope would make it affine, a linear function plus a constant. But choosing zero control gives $\Phi_\gamma(t,x) \geq \mathbb{E}|x + B_1 - B_t| \geq |x|$: the average absolute value is at least the absolute value of the average. This rules out an affine value: domination of $|x|$ in the positive tail requires slope at least one, while the negative tail requires slope at most minus one. The heat identity and the needed regularity are in OpenAI (2026a), Lemmas 2.2–2.3.

Thus q' is strictly increasing, and right contact gives $q'(t) < q'(s) \leq 1$ for $0 < t < s$. Symmetry supplies $q(0) = 0$, while contact gives $q(s) = s$. Integrating again gives too little area. Strictness was needed only at positive times; symmetry can make the third spatial derivative vanish at $(0,0)$.

Although the forward path is Brownian here, the backward value receives data $\Phi_\gamma(s,\cdot)$ from the later coefficient. The formula for the globally zero candidate cannot be reused on this initial interval. The case where zero itself is supported was already included in Section 5.3, using its atom and the one-sided slope bound.

Empty support. This would mean $\gamma = 0$ everywhere. The zero candidate fails the first-variation test:

$$D\mathcal{P}(0)[1] = \frac{1}{4} - \frac{1}{\pi} < 0.$$

The direction adds an atom at zero and is admissible. Section 7.1 computes the derivative from the explicit Gaussian response; no minimiser identity is used. A minimiser therefore has nonempty support.

Terminal gap. Now suppose a maximal missing interval is $(r,1)$. Its left endpoint $r < 1$ is supported, and its constant height is positive, by the same mass argument as in Section 5.3. We have $q(r) = r$ and $q'(r) \leq 1$, and q' is strictly convex on $(r,1)$. There is no contact condition at one. We use the bound $q \leq 1$ instead. We will show that q stays below the diagonal throughout the gap, which makes the tail cost $G(r)$ negative. If it reached the diagonal, the area comparison would force its slope above one; convexity would preserve enough of that excess to violate $q \leq 1$.

Put $h(t) = q'(t) - 1$, so $h(r) \leq 0$. If $q(t) \geq t$ at some $r < t < 1$, strict integration below the chord on the finite interval $[r,t]$ would give

$$0 \leq q(t) - t = \int_r^t h(v)\,dv < \frac{t-r}{2}(h(r) + h(t)) \leq \frac{t-r}{2}h(t).$$

Hence $h(t) > 0$. To control a later slope, use the same chord rule on the larger interval $[r,s]$, where $t < s < 1$. At its interior point t this gives

$$h(t) \leq \frac{s-t}{s-r}h(r) + \frac{t-r}{s-r}h(s).$$

Solving for the right endpoint gives

$$h(s) \geq h(t) + \frac{s-t}{t-r}(h(t) - h(r)) \geq h(t) > 0.$$

Consequently $q(s) \geq t + (s-t)(1 + h(t))$. Choose any time

$$t + \frac{1-t}{1+h(t)} < s < 1.$$

Such a choice exists because $h(t) > 0$, and the lower bound now gives $q(s) > 1$, contradicting $q(s) \le 1$ strictly before time one. Thus $q(t) < t$ throughout $(r, 1)$, giving $G(r) = \int_r^1 (q(t) - t)\, dt < 0$, contrary to the addition test. Only finite-interval convexity and the bound on q were used; no value for $q'(1)$ was assumed.

The source's different terminal argument proves the stronger estimate $1 - q(t) \ge C\sqrt{1-t}$ near one for any finite constant terminal coefficient, without the left contact data used here. Its proof is OpenAI (2026a), Lemma 6.3 and is retained in the *Long Reasoning Record*, §5, terminal estimate. Auffinger–Chen–Zeng, Remark 7 had already recorded the absence of terminal plateaus.

These cases exhaust the complement of the support. Contact now gives $q(t) = t$ for every interior time, and symmetry includes zero. Differentiating gives $\mathbb{E}[a(t, X_t)^2] = 1$, completing the theorem.

5.5 Two checks on the gap argument

Question 3: tilt the chord. Suppose a differentiable function f on $[2, 5]$ has $f(2) = 2$, $f(5) = 5$, $f'(2) = 1/2$ and $f'(5) = 3/2$. If f' is strictly convex, can these data hold? What changes if strict convexity is weakened to convexity? The endpoint chord is now tilted, so a pointwise comparison with the constant one is unavailable.

Question 4: the missing right endpoint. In the terminal proof, why does a point with $q(t) \ge t$ force $h(t) > 0$? Which inequality keeps the later slopes above one, and where does the bound $q \le 1$ enter? Explain the two stages without assigning any value to $q'(1)$.

Answers are in Section 7.4.

6. Return to the spins

We can now return to the arrangement of nearly optimal spin choices. A separate theorem of Auffinger–Chen–Zeng, Proposition 1, equation (5) applies at each point in the minimiser's support. Combining it with full support: for every fixed overlap $r \in [0,1)$ and tolerances $\varepsilon, \eta > 0$, with probability over the couplings tending to one exponentially fast as $n \to \infty$, there exist configurations σ, τ with

$$|R(\sigma,\tau) - r| < \varepsilon, \qquad H_n(\sigma)/n,\ H_n(\tau)/n \geq P_* - \eta.$$

If k sites differ, $k/n = (1-R)/2$. The overlap tolerance therefore gives

$$\left|\frac{k}{n} - \frac{1-r}{2}\right| < \frac{\varepsilon}{2}.$$

For example, fix an energy loss $\eta > 0$ per spin and take $r = 1/2$ and $\varepsilon = 0.02$. With probability tending to one exponentially fast, the same instance has two configurations with energies per spin at least $P_* - \eta$ and a differing-site fraction between 0.24 and 0.26. No numerical threshold for n is asserted. At target zero, the differing-site fraction can be arbitrarily close to one half. The pair may depend on the chosen target and the couplings. This is existence for each fixed target, not a statement about typically sampled pairs or a procedure for finding them. The companion *Long Reasoning Record*, §9.3, "Near-ground configurations at a prescribed overlap", gives the probability bound and the covariance adjustment.

7. Optional calculations

The no-gap argument and the return to spins are complete. Section 7.1 checks the zero-candidate descent by Gaussian calculus using the stated moment identity. Section 7.2 explains the value equation and stochastic chain rule behind that identity. Section 7.3 describes the mechanism of the strict-curvature estimate; its full proof remains in the sources. Section 7.4 gives the answers to the earlier questions.

7.1 Why the zero candidate does not minimise

Return to the Gaussian value and slope from Section 3. With $\gamma = 0$, the state is B_t. Differentiating the slope once more gives

$$X_t = B_t, \qquad u_0(t,x) = 2F_{\mathrm{G}}\left(\frac{x}{\sqrt{1-t}}\right) - 1, \qquad a_0(t,x) = \sqrt{\frac{2}{\pi(1-t)}}\, e^{-x^2/(2(1-t))}.$$

A Gaussian integral gives $\mathbb{E}e^{-B_t^2/(1-t)} = \sqrt{(1-t)/(1+t)}$. Substituting into $q_0' = \mathbb{E}[a_0(t,B_t)^2]$ and integrating from zero yields

$$q_0'(t) = \frac{2}{\pi\sqrt{1-t^2}}, \qquad q_0(t) = \frac{2}{\pi}\arcsin t.$$

This recovers the half-time value 1/3. It also shows that $q_0(t) \to 1$ while $q_0'(t) \to \infty$: convergence of a function at one does not bound its derivative there. The constant direction $\eta = 1$ is admissible and adds an atom at zero. Its cost is

$$D\mathcal{P}(0)[1] = \frac{1}{2}\int_0^1 \left(\frac{2}{\pi}\arcsin t - t\right) dt = \frac{1}{4} - \frac{1}{\pi} < 0.$$

So a small positive constant coefficient improves the zero candidate. This calculation uses the heat and variation formulas, not a minimiser hypothesis; see also Chen–Handschy–Lerman, proof of Theorem 6.

7.2 The value equation and the moment identity

At $t = 1$ there is no noise or response left to accumulate, so the value is $|x|$. At earlier times it includes the best remaining response. Its partial differential equation (PDE) runs backward from that final value:

$$\partial_t \Phi_\gamma = -\frac{1}{2}\left(\partial_{xx}\Phi_\gamma + \gamma(t)(\partial_x \Phi_\gamma)^2\right), \qquad \Phi_\gamma(1, x) = |x|.$$

With the spatial derivatives u and a defined in Section 3.3, the equation expresses a recursive choice: choose a response for a short first step, then use the best continuation from the state reached. At a continuity point of γ, this gives the formal first-order relation

$$\Phi_\gamma(t, x) = \sup_{|v|\le 1}\left\{\mathbb{E}\Phi_\gamma(t + h, x + \gamma(t)vh + \sqrt{h}Z) - \tfrac{1}{2}\gamma(t)v^2 h\right\} + o(h).$$

Subtract the present value, divide by h, and Taylor-expand. The identities $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = 1$ give

$$0 = \Phi_t + \tfrac{1}{2}\Phi_{xx} + \sup_{|v|\le 1}\gamma(t)(v\Phi_x - v^2/2).$$

The square completion from Section 3.3 gives a supremum of $\gamma u^2/2$, attained at $v = u$, and hence the stated PDE. This calculation explains the equation; justifying the recursion and the Taylor expansion requires the control representation and regularity results. See Auffinger–Chen, Theorem 2 and Corollary 2 for the zero-temperature control formula and Jagannath–Tobasco, Lemma 18 for the control method with finite-temperature terminal data. The extension after Auffinger–Chen's Corollary 2 covers integrable coefficients. The regularity used here is in OpenAI (2026a), Lemmas 2.1–2.2.

Itô's rule is the chain rule for Brownian paths. Along $dX_t = b_t\,dt + dB_t$, the coefficient of dt (the **drift**) in a smooth $f(t, X_t)$ is $f_t + b_t f_x + f_{xx}/2$, and its noise term is $f_x\,dB_t$. Differentiate the PDE in x and take $b_t = \gamma u$. The drift of the slope cancels:

$$du(t, X_t) = \underbrace{(u_t + \gamma u u_x + u_{xx}/2)}_{0}\,dt + a(t, X_t)\,dB_t.$$

Thus $M_t = u(t, X_t)$ is a bounded **martingale**: its conditional expected later value, given observations up to time t, is its present value. Zero field gives symmetry and $M_0 = u(0, 0) = 0$. Applying Itô's rule once more to M_t^2 gives

$$d(M_t^2) = 2M_t a(t, X_t)\, dB_t + a(t, X_t)^2\, dt.$$

The noise term has mean zero because its coefficient is nonanticipating and bounded on each strip ending before one. Taking expectations and integrating yields $q(t) = \int_0^t \mathbb{E}[a(s, X_s)^2]\, ds$, which gives the asserted derivative. Since $\mathbb{E}M_t = 0$, $q(t)$ is also the variance of the slope response.

For each fixed $T < 1$, the required spatial derivatives are bounded on $[0, T] \times \mathbb{R}$, and step-coefficient and smooth-terminal approximations justify the identities. The bounds depend on the coefficient and strip. The PDE holds almost everywhere in time. These are not bounds up to time one.

7.3 Where the strict-curvature input comes from

The proof in Section 5 used strict curvature on a positive constant interval. This calculation explains how that expectation acquires a sign; the full analytic proof remains in the cited sources. Take any admissible coefficient with $\gamma = c > 0$ on an open interval, without assuming minimality. Set $b = \Phi_{xxx}$ and $d = \Phi_{xxxx}$ there. Differentiating the PDE and applying Itô's rule gives

$$q'' = \mathbb{E}[b^2 - 2ca^3], \qquad q''' = \mathbb{E}[d^2 - 12cab^2 + 6c^2a^4].$$

The negative middle term prevents a pointwise positivity argument. The source instead averages with the spatial probability density $f_t(x)$ of X_t at positive time. As in the half-time example, for any integrable function F, grouping paths by their reached position gives

$$\mathbb{E}[F(X_t)] = \int_{\mathbb{R}} F(x) f_t(x)\, dx.$$

These are probability weights on states x, distinct from the Stieltjes weights $\mu_\gamma(ds)$ on overlap scales. For step coefficients and smoothed terminal data, this density factors into positive functions propagated from the two ends of the time interval. Bounds on their logarithmic derivatives supply the signs needed in integration by parts.

The operation can be written directly for the positive density f_t. For a smooth function F with integrable terms and vanishing boundary product Ff_t,

$$\mathbb{E}\left[F'(X_t) + F(X_t)\frac{f_t'(X_t)}{f_t(X_t)}\right] = \int_{\mathbb{R}} (Ff_t)'(x)\, dx = 0.$$

For a small example of this operation, let Z be standard normal and $F(x) = 2x$. Its density has logarithmic derivative $-x$, so $\mathbb{E}[2 - 2Z^2] = 0$. Although $2x^2 - 1$ is negative near zero, adding that zero-mean correction gives

$$(2x^2 - 1) + (2 - 2x^2) = 1.$$

Hence $\mathbb{E}[2Z^2 - 1] = 1 > 0$. The boundary term vanishes because a polynomial times the Gaussian density tends to zero. This illustrates how a correction changes the visible sign without changing the expectation; it supplies none of the SK-specific sign estimates.

At the step-coefficient, smooth-terminal stage, write $R_c = d^2 - 12cab^2 + 6c^2a^4$. The source constructs a correction C from the spatial derivatives and the density factors, with

$$\mathbb{E}[C(t, X_t)] = 0, \qquad R_c(t, x) + C(t, x) \geq \frac{c^2}{1000} a(t, x)^4.$$

A rational polynomial identity gives the second inequality by writing the remainder as a sum of nonnegative terms. The density bounds and vanishing boundary products justify the first. Consequently

$$q'''(t) = \mathbb{E}[R_c(t, X_t)] = \mathbb{E}[(R_c + C)(t, X_t)] \geq \frac{c^2}{1000} \mathbb{E}[a(t, X_t)^4].$$

This is how the correction produces a useful lower bound for the expectation even though R_c can be negative at particular states.

To reach an arbitrary admissible coefficient, fix a time inside the plateau and approximate in L^1 by step coefficients, keeping the same value c near that time. Uniform convergence of spatial derivatives through order four on a fixed strip below one, together with convergence of the coupled paths, passes the quantitative inequality to the limit. Strictness is then established for the limit itself: the constant-coefficient heat formula expresses a as a nonnegative mean plus c times a strictly positive variance of a nonconstant slope under a positively weighted Gaussian law. Hence $a(t, x) > 0$ and its fourth moment is positive. Positivity of the approximants alone would not suffice. The auxiliary logarithmic derivatives need not converge: they have already been removed by integration.

The analytic details are OpenAI (2026a), §§3–5 and the companion *Long Reasoning Record*, §4.4, "How the signs make a correction with zero expectation useful", and §4.5, "A fixed-time inequality and its plateau application".

7.4 Answers to the questions

Question 1. Adding mass at $1/2$ gives cumulative heights $0, 2, 2 + \varepsilon, 3 + \varepsilon$, in their successive intervals; removing it at $1/4$ gives heights $0, 2 - \varepsilon, 3 - \varepsilon$. Both functions remain nonnegative and nondecreasing. Removing mass at $1/2$ instead lowers the height from 2 to $2 - \varepsilon$ there, creating a downward jump, so it is forbidden even though all values remain nonnegative. Scaling both weights preserves all admissibility conditions. These tests concern allowed candidates; a smaller ordinary area alone does not establish a smaller Parisi value.

Question 2. The state has accumulated variance $1/4$, so it is distributed as $Z/2$. The remaining noise has variance $3/4$, and its standard deviation in the value-slope formula is $\sqrt{3}/2$. Thus

$$q_0(1/4) = \int_{\mathbb{R}} \left(2F_{\mathrm{G}}(z/\sqrt{3}) - 1\right)^2 \varphi(z)\, dz.$$

The two variances agree at half time, but not here. For nonzero z, the magnitude of the response is now smaller, so $q_0(1/4) < 1/3$. The arcsine formula in Section 7.1 also gives this comparison.

Question 3. A strictly convex f' lies below its endpoint chord. That chord has mean height $(1/2 + 3/2)/2 = 1$, hence integral three. Therefore $f(5) - f(2) = \int_2^5 f'(t)\,dt < 3$, contradicting the prescribed values. With non-strict convexity the data are possible: $f(t) = 2 + (t-2)/2 + (t-2)^2/6$ has affine derivative and satisfies all four endpoint values. The argument needs a strict loss of area; the common endpoint upper bound of one in Figure 2 is convenient, not essential. This is a calculus example, not a candidate SK moment function.

Question 4. Since $q(r) = r$, reaching or passing the diagonal at t means $\int_r^t h \geq 0$. Strict convexity places h below its endpoint chord, so that integral is less than $(t-r)(h(r)+h(t))/2$. With $h(r) \leq 0$, this forces $h(t) > 0$. The chord rule on $[r, s]$, solved for $h(s)$, keeps every later value at least $h(t)$. Integrating gives $q(s) \geq t + (s-t)(1+h(t)) > 1$ for a suitable $s < 1$, violating the bound. If q never reaches the diagonal, then $G(r) < 0$ instead. Thus both possibilities contradict the conditions on a minimiser.

Further reading

Full support alone does not give a density. Additional differentiation of the identities $q = t$ and $\mathbb{E}[a^2] = 1$, followed by another use of the integrated curvature estimate on locally constant comparison coefficients, proves that γ is smooth, $\gamma(0) = 0$, and $\mu_\gamma = \gamma'(t)\,dt$ with $\gamma'(t) > 0$ for $0 < t < 1$ (OpenAI (2026a), Corollary 6.4). The *Long Reasoning Record*, §§9.1–9.2, supplies this additional argument; Chen's earlier theorem had already given smooth nonnegative density and full support.

Further martingale arguments recover the ground-state value from the terminal slope response. They use the first curvature moment $\mathbb{E}[a]$, distinct from the second moment $\mathbb{E}[a^2] = 1$ used here. See OpenAI (2026a), Corollary 7.1 and the *Long Reasoning Record*, §6.

The separate circuit theorem in OpenAI (2026b) resolves the eventual-optimality conjecture for the Quantum Approximate Optimization Algorithm (QAOA) in Basso–Farhi–Marwaha–Villalonga–Zhou, §7, equation (7.2). The *Long Reasoning Record*, §8, develops its circuit construction and the order of limits. It requires additional arguments beyond full support.

Sources and scope of verification

This account reconstructs selected parts of OpenAI's analytic manuscript. The global scaling argument for contact follows Chen's variational proof; the terminal convexity argument in Section 5.4 is an ordinary deduction from the stated curvature, contact and boundedness inputs. It is not a claim that this alternative proof has been replayed in Lean. The Lean implementation, challenge statement and Comparator configuration concern a minimiser's support and moment identities. They assume a minimiser; they do not prove the external Parisi formula or attainment. The separate scalar implementation and statement concern the value identity and finite Gaussian approximation. Neither selected statement covers positive density, the external near-ground overlap theorem used in Section 6, or the complete QAOA circuit theorem. The implementations use Lean 4 (de Moura–Ullrich, 2021) and mathlib (The mathlib Community, 2020).

Comparator acceptance and research-v2 extraction remain unknown; see the *Long Reasoning Record*, §13. Kernel checking establishes a declared statement under its definitions and axioms. Its correspondence with the analytic theorem, independent proof review and reader comprehension are separate questions. The links below select the editions used.

- **OpenAI (2026a).** *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*, 27 September 2026. Pinned manuscript. §2 supplies regularity, Proposition 5.1 the strict curvature estimate, §6 the support proof, and Corollary 6.4 the additional density conclusion.
- **Hong-Bin Chen (2026).** *FRSB in the SK spin glass: convergence to full-interval support at zero temperature*. Version 1, Theorem 1.3 for earlier full support and smooth nonnegative density; Proposition 3.2 for contact by scaling the coefficient. Strictly positive density in the further-reading note uses the separate OpenAI corollary.
- **Antonio Auffinger and Wei-Kuo Chen (2017).** *Parisi formula for the ground state energy in the mixed p-spin model*, Annals of Probability 45(6B), 4617–4631. Version 2. The variational formula and attainment used in Section 3.2.
- **Aukosh Jagannath and Ian Tobasco (2016).** *A dynamic programming approach to the Parisi functional*, Proceedings of the American Mathematical Society 144, 3135–3150. Version 4. The control method, with finite-temperature terminal data; it is not by itself the zero-temperature regularity theorem used here.
- **Wei-Kuo Chen, Madeline Handschy and Gilad Lerman (2018).** *On the energy landscape of the mixed even p-spin model*, Probability Theory and Related Fields 171, 53–95. Version 4. Proposition 4 gives the first-variation formula used in Section 5.1; the proof of Theorem 6 gives the zero-candidate descent.
- **Antonio Auffinger, Wei-Kuo Chen and Qiang Zeng (2020).** *The SK model is infinite step replica symmetry breaking at zero temperature*. Version 2, Proposition 1 and Remark 7. Near-ground overlap pairs and the earlier terminal-plateau observation.
- **OpenAI (2026b).** *QAOA attains the SK ground-state energy in the thermodynamic-first limit*, 25 September 2026. Pinned manuscript. The separate circuit theorem.
- **Joao Basso, Edward Farhi, Kunal Marwaha, Benjamin Villalonga and Leo Zhou (2022).** *The Quantum Approximate Optimization Algorithm at High Depth for MaxCut on Large-Girth Regular Graphs and the Sherrington-Kirkpatrick Model*. Full preprint, version 3, §7, equation (7.2). The eventual-optimality conjecture.